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Real-Time Wireless Monitoring for Three Phase Motors in Industry: A Cost-Effective Solution using IoT

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Abstract— **In recent days modern environment industries are facing rapid flourishing for performance capabilities and their requirements for corporate clients and industrial sector. Internet of Things (IoT) is an innovative and rapidly growing field for automation and evaluation in networks, Artificial Intelligence, data sensing, data mining, and big data. These systems have a great tendency to monitor and control different process used in industries. IoT systems have been implemented and have applications in different industries due to their cost-effectiveness and flexibility In this paper we have developed a system which includes real-time monitoring of current reading of three-phase motor through a wireless network. With the help of this system, data can be saved and monitored and then transmitted to cloud storage. This system contains Arduino-UNO board, ACS-712 current sensor, ESP-8266 Wi-Fi module which sends information to an IoT API service THING-SPEAK that behave like a cloud for various sensors to monitor data. The proposed system was successfully deployed in Aisha Steel Mills, Karachi, Pakistan.**

Keywords— IOT (Internet of Things), API (Application Program Interface), wireless monitoring, three phase motor

I. INTRODUCTION

Idea behind this research is to design a wireless monitoring system of three phase motors in industry with the help of IoT based system. . In [1-2] authors recorded and captured different environmental factors like humidity, light intensity, temperature and several other parameters were recorded, measured and then information was sent wirelessly to Thing-Speak with the help of Arduino UNO, microcontroller and sensor networks. This work focuses considerably on MATLAB visualization and analysis. The data was sent to Thing-Speak cloud. In contrast with sensor LPC2148, Arduino UNO placed in this system is easy to install and cost effective. Researchers in [3] described an IoT based real-time weather monitoring system comprised of Raspberry Pi which was not user friendly, complex in contrast with Arduino due to Python language and its complex operating system. In [4] authors described an Arduino based monitoring system. Authors imported information from various sensors to excel which is cumbersome when compared to Thing-Speak. Authors in [5] proposed and presented a monitoring system operated wirelessly for climate monitoring composed of Radio and Raspberry Pi. They suggested Zigbee module for collection of data from various sensor based networks at a base station (Raspberry Pi). That model can be enhanced huge scale implementation, however in current scenario, cloud connectivity is absent. The primary and essential purpose of wireless readings and monitoring of 3 phase motor is to capture the parameters of current that are drawn by induction motor [6, 7]. Several kinds of motor faults known as rotor fault, short winding fault, air gap fault, bearing load faults e.tc can be observed using this technique [8, 9]. In case of any indiscretion appear in induction motors can cause immense downtime and revenue in terms of maintenance and business loss as well. This study involves a real time wireless monitoring of 3 phase induction motors through different parameters like value of 3 phase voltages and current, speed of the shafts and torque [10-12]. Based on these parameters, output power and efficiency was calculated. Wireless monitoring is also very effective and economical. Sensors monitors the parameters of induction motor and deliver the information through Wi-Fi module and it will be saved at cloud storage which will be accessed by end user. This system contains Arduino UNO board, ASC 712 current sensor, ESP-8266 Wi-Fi module which sends result to an IoT API service THINGSPEAK that behave as a cloud for various sensor to monitor data [13, 14]. With the advent of Internet and wireless technologies, communication is much more flexible between human as well as among devices without any interference of humans to control communication. This concept is termed as Internet of Things (IoT). As industries are progressing day by day there is a need of cost effective solution to monitor and manage daily power. In order to reduce maintenance cost, human errors and faults and labour cost with more effective and efficient manner for power management system, a solution is implemented for power and energy related issues.

This paper is mainly focused on energy and power monitoring using IoT based monitoring system. The purpose of this study is to propose and implement a low cost wireless sensor based network which will be efficient and competent of getting automatic results and readings. Due to this system, the users will be informed and updated of his electricity usage and users will be capable to reduce power loss, power wastage and consumption cost of electricity. In this research we tried to explore some cost effective techniques like use of Ardiuno board and Raspberry Pi boards.

II. PROBLEM STATEMENT

Induction motor (IM) is one of the key factors and plays an important role in most industries. Hence it requires a lot of monitoring and controlling. Stator coil current is preferably monitored as the stator coil current keeps principle discriminative features that are essential for detection and classification of errors within induction motor. Wired system occupies a lot of space and maintenance therefore solution of this problem was needed by the industry. The induction motor requires or should have to be protected to minimize possible problems that are mentioned as high voltage, high current, extreme temperature etc as it is widely used motor in industry. In Aisha Steel Mills located in Karachi these induction motor are used for cooling purpose and they are backbone of their entire system. So, the protection of these motors is a major problem due to excessiveness readings of current. Their system consisted of 14 3 phase motors of 5.5KW load and 11 3 phase motors of 12.5 KW are used for cooling purpose. The main task was to monitor the current reading of 3 phase motors of entire plant and transmit those readings via Wi-Fi to provide graphical display of real time readings for current values of motors wirelessly because managing a wired system and gathering of data from that wired network of these motors is not an easy task. One of the main reason for going towards a wireless sensor based monitoring system is the maintenance cost which is required for wired network and it is relatively high as compared to wireless sensor network.

III. DESIGN AND METHODOLOGY

The main and controlling unit is a microcontroller (Arduino UNO) and behaves as the central unit for the whole system, its interfaces which connected with the sensor chip at the input end for gathering current readings and interfaces which connected with the wireless module at the output to transmit the sensed information towards cloud using Internet. The microcontroller polls the sensor to retrieve information and transmits using the Internet to Thing-Speak application.

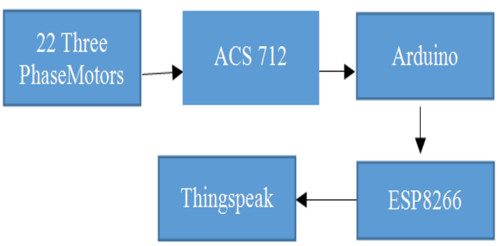

Fig. 1. Designing methodology of 3 phase monitoring system

Fig.1 shows the main block diagram of the wireless monitoring system. Current rating and voltages readings of twenty motors was observed using ACS712 and transmitted to the Thinspeak cloud server using ESP8266.

A. Main Controller

Fig. 2. Arduino-UNO microcontroller board

The central hardware component is microcontroller which interconnects other components of the system. Arduino-UNO microcontroller served our purpose well because of ease in use, robustness and cost effectiveness. Figure 2 demonstrated a picture of Arduino UNO microcontroller used for controlling purpose in our system. Microcontroller board is comprise on the ATmega328P with 14 input/output pins, 6 analog input pins, a USB connection, 16 MHz quartz crystal, a power jack, and a reset button. Power will be provided through a battery. It is programmable with the Arduino IDE (Integrated Development Environment) via a type B USB cable.

B. ACS-712

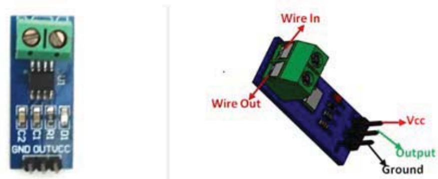

Fig. 3. ACS712 (Current Sensor module)

ACS-712 is a Hall Effect based fully integrated lined current sensor with 2.1 KV-RMS voltage isolation with least resistance current conductor and having ability to measure current AC or DC ranging from -5A to -30A and +5A to 30A.Its provides cost-effective and accurate solution for both type of current in industrial marketable with other communication systems [15-17]. The sensing element module enable simple for implementation with kind of typical application include controlling purpose, load detection and management also suitable for switch mode power supply and over current failure protection. We need to choose the correct range for our undertaking since we need to exchange off exactness for higher range modules. The resulted analog voltage of the component depends on the current passing thought out the wire, hence it is very easy to interconnect this module with any microcontroller ,Since the sensor followed the hall-effect principle so some description regarding the hall effects based sensor as the ACS-712 are follows: The Hall impact is created by the voltage distinction (the Hall voltage) present across an electrical conductor, which transverse to an electrical current in the conductor and to an applied magnetic field is perpendicular to the current [18-20]. According to the principle of Hall Effect which was discovered by Dr. Edwin Hall in 1879 stated as “When a current carrying conductor is placed into the magnetic field, a voltage is generated at the edges which were perpendicular to the directions of both the current and the magnetic field [21-22]. For demonstration a figure is given below which describe the working of this principle. A small sheet of semiconductor material would be known as (Hall Element) is carrying a current I and is placed into the magnetic field B which is perpendicular to the direction of current flow. Current will not be uniform in this case due to the presence of Lorentz force and will cause voltage across its edges which are perpendicular to the direction of current and field.

C. *Wi-Fi Module*

Fig. 4. ESP8266-Wi-Fi Module

For sending sensed data through current sensor ACS712 towards the open source cloud ThingSpeak, Arduino UNO interfaces at the output terminal having Wireless interface ESP8266 demonstrated in (Figure 4). It is a cost effective Wireless microchip with complete OSI layer or TCP/IP stack. It operates on the 3.3V that is supplied by Arduino UNO. The module is configured through AT commands and requires the pattern to be represented as end user. The module can be pursuing as end user and server. It obtains an IP on being connected to Wi-Fi through the module and then broadcast over the Internet.

D. *Fundamental Block Diagram*

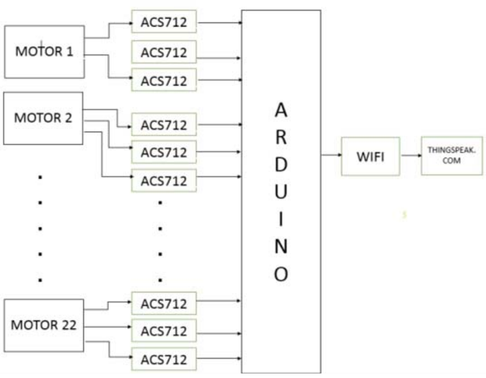

Fig.5. Block Diagram for 3 phase monitoring system

Figure 5 represents the block diagram for this research. The diagram describes the basic operation mode of the structure with functionality where ACS712 gives live readings of motors current synchronized with the microcontroller which transmits the data using Wi-Fi module through the Internet to the Thing-Speak IoT app.

E. *Software Implementation*

PCB schematic was designed using Proteus and thingspeak cloud service was used for IoT application. Proteus software is a proprietary software gadget which is mainly used for electronic circuit architectures and their simulation and automation. This tool is designed mainly for electronic engineers and technicians to perform real time scenarios and schematics for circuit designing. ThingSpeak is an open-source Internet of Things (IoT) application and API to capture information from different gadgets using the HTTP protocol using Internet or within a Local Area Network. Due to this we can get status of different apps, tracking, sensor based applications etc in graphical and visual form. After through testing of every component and integrated them according to given diagram. Initially, the Wi-Fi shield was initialized by giving AT commands in the required pattern to make it work as a client. ACS712 which provides current results in real-time. Once information is sensed by the microcontroller and sent to the cloud and then IoT analytics service of ThingSpeak to accumulate, envision and analyse real time data. ThingSpeak produces instant graphical results of the real-time data uploaded by our system.

IV. EXPERIMENTAL RESULTS

This Section explains the experimental results of the system and shows sensor data available to the end user using the mobile application and the graphical results of motors current reading.

created_at	entry_id	field1	field2	field3
2019-06-17 21:08:57 PKT	1	1.2	1.3	1.2
2019-06-17 21:09:15 PKT	2	1.2	1.3	1.2
2019-06-17 21:09:33 PKT	3	1.3	1.3	1.2
2019-06-17 21:09:51 PKT	4	1.2	1.3	1.2
2019-06-17 21:10:09 PKT	5	1.2	1.3	1.2
2019-06-17 21:10:27 PKT	6	1.3	1.3	1.3
2019-06-17 21:10:45 PKT	7	1.3	1.3	1.2
2019-06-17 21:11:03 PKT	8	1.2	1.3	1.3
2019-06-17 21:11:21 PKT	9	1.2	1.3	1.3
2019-06-17 21:11:39 PKT	10	1.2	1.3	1.2
2019-06-17 21:11:57 PKT	11	1.2	1.3	1.2
2019-06-17 21:12:15 PKT	12	1.3	1.3	1.3
2019-06-17 21:12:33 PKT	13	1.3	1.3	1.2
2019-06-17 21:12:51 PKT	14	1.3	1.3	1.3
2019-06-17 21:13:09 PKT	15	1.2	1.2	1.2
2019-06-17 21:13:27 PKT	16	1.2	1.3	1.2
2019-06-17 21:13:45 PKT	17	1.3	1.3	1.3
2019-06-17 21:14:03 PKT	18	1.2	1.3	1.3
2019-06-17 21:14:21 PKT	19	1.2	1.3	1.3
2019-06-17 21:14:39 PKT	20	1.3	1.3	1.2

Fig. 6. Data feed Values of three phase motors

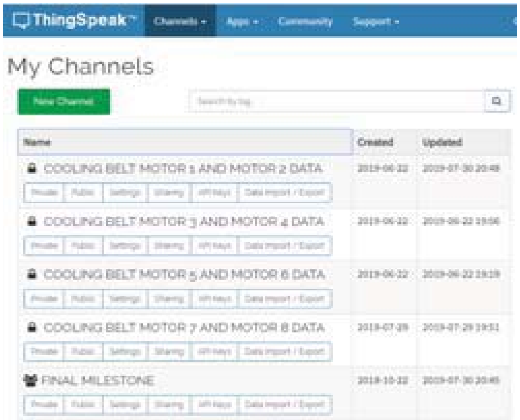

Fig. 7a. Channels for the motor data at Things Speak

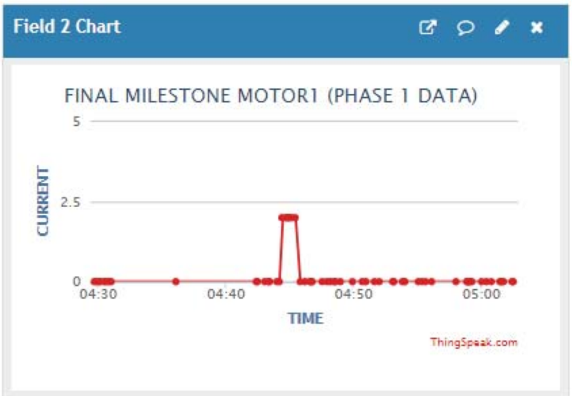

Fig. 7b. Graphical Analysis of Motor current rating for Phase 1

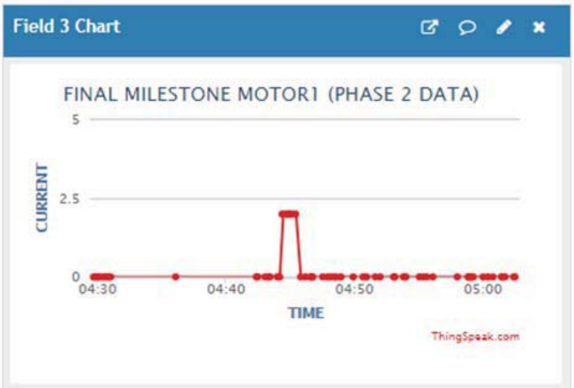

Fig. 7c. Graphical Analysis of Motor current rating for Phase 2

Fig. 7d. Graphical illustratuib of Motor current rating for Phase 1 and 2

Fig. 7e. Graphical illustratuib of Motor current rating for Phase 1 and 2

Fig. 7a. demonstrated that how the channels were created in the Things speak a cloud service for data logging and monitoring. Fig. 7b and 7c shows the results of the real time parameters related to the three phase motors deployed in the industry. Data can be logged for the further analysis and can also be imported easily from the cloud server. Current parameters of induction motors that are used in cooling belts of BAF (Control Room) have been recorded and analyzed successfully in Aisha Steel Mills. These current parameters will be helpful in the future to avoid the destruction of motors in terms of excess of current. The system has been tested in the industry and the Motor's current parameters have been tested and calculated per batch of the product, moreover data monitoring on the cloud level showed the utilization and load consumption by each motors.

V. DEPLOYMENT OF SYSTEM IN INDUSTRY

Fig. 8. Glimpse of installation in industry

Wireless controlling module was connected in the main controlling panel of the motors. The main current wires were connected to the ACS-712 and module transmitted the data to the Things Speak cloud service. The proposed solution worked very well in industry in a very cost effective manner with less wired system.

VI. CONCLUSION

The proposed solution for the industry worked very well with the three phase motors, initially it was tested up to three days, and the module was connected in the main controlling panel of twenty two motors of Aisha steel mills situated in Karachi, Pakistan. Industrialists satisfied with the work as it has wireless and cost effective solution with less maintenance. Data logging, monitoring and analysis are another advantage of the system.

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